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7.2.6 Elastic energy

When a body is stretched, work is done against the restoring force (internal force). This work done is stored in the body in the form of elastic energy. Consider a wire whose un-stretch length is L and area of cross section is A. Let a force produce an extension l and further assume that the elastic limit of the wire has not been exceeded and there is no loss in energy. Then, the work done by the force F is equal to the energy gained by the wire.

The work done in stretching the wire by dl,

$dW = F\,dl$

The total work done in stretching the wire from 0 to l is

$$W = \int_0^l F\,dl \qquad (7.11)$$

From Young's modulus of elasticity,

$$Y = \frac{F}{A}\times\frac{L}{l} \Rightarrow F = \frac{YAl}{L} \qquad (7.12)$$

Substituting equation (7.12) in equation (7.11), we get

$$W = \int_0^l \frac{YAl}{L}dl$$

Since, l *is* the dummy variable in the integration, we can change l *to* l'(not in limits), therefore

$$W = \int_0^l \frac{YAl'}{L}dl' = \frac{YA}{L}\left(\frac{l'^2}{2}\right)\Big|_0^l = \frac{YA}{L}\frac{l^2}{2} = \frac{1}{2}\left(\frac{YAl}{L}\right)l = \frac{1}{2}Fl$$

$$W = \frac{1}{2}Fl = \textit{Elastic potential energy}$$

Energy per unit volume is called energy density, $u = \dfrac{\text{Elastic potential energy}}{\text{Volume}} = \dfrac{\frac{1}{2}Fl}{AL}$

$$\frac{1}{2}\frac{F}{A}\frac{l}{L} = \frac{1}{2}\,(\text{Stress}\times\text{Strain}) \qquad (7.13)$$

EXAMPLE 7.5

A wire of length 2 m with the area of cross-section $10^{-6}m^2$ is used to suspend a load of 980 N. Calculate i) the stress developed in the wire ii) the strain and iii) the energy stored. Given: $Y = 12\times10^{10}N\,m^{-2}$.

Solution

(i) $\text{stress} = \dfrac{F}{A} = \dfrac{980}{10^{-6}} = 98\times10^{7}N\,m^{-2}$

(ii) $strain = \dfrac{stress}{Y} = \dfrac{98\times10^{7}}{12\times10^{10}} = 8.17\times10^{-3}$ (no unit)

(iii) Since, volume $= 2\times10^{-6}m^3$

$Energy = \frac{1}{2}(stress\times strain)\times volume \Rightarrow$

$\frac{1}{2}(98\times10^{7})\times(8.17\times10^{-3})\times2\times10^{-6} = 8\,joule$

7.2.7 Applications of elasticity

The mechanical properties of materials play a very vital role in everyday life. The elastic behavior is one such property which especially decides the structural design of the columns and beams of a building. As far as the structural engineering is concerned, the amount of stress that the design could withstand is a primary safety factor. A bridge has to be designed in such a way that it should have the capacity to withstand the load of the flowing traffic, the force of winds, and even its own weight. The elastic behavior or in other words the bending of beams is a major concern over the stability of the buildings or bridges. For an example, to reduce the bending of a beam for a given load, one should use the material with a higher Young's modulus of elasticity (Y). It is obvious from Table 7.1 that the Young's modulus of steel is greater than aluminium or copper. Iron comes next to steel. This is the reason why steel is mostly preferred in the design of heavy duty machines and iron rods in the construction of buildings.